AQ4.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times 3}(R)$M3×3 (R)) in column A with their bases in column B in Table : M2W4AQ2.

Choose the correct option.

- [ ] a → → ii
- [ ] a → → i
- [ ] b → → iii
- [ ] c → → i
- [ ] b → → ii
- [ ] c → → iii

No, the answer is incorrect.

Score: 0

Accepted Answers:

a → → ii
b → → iii
c → → i

If SS be a subset of R^5 R5 such that $span(S) = R^5$ span(S) = R5, then what is the minimum number of elements possible in SS?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

1 point

If S_1 S1 is a maximal linear independent set and S_2 S2 is a minimal spanning set of a vector space VV , then which of the following option(s) is (are) true?

- [ ] For any $v \in S_1$ v ∈ S1, $S_1 \setminus \{v\}$S1 \ {v} is a linearly independent.
- [ ] For any $v \in S_2$ v ∈ S2, $S_2 \setminus \{v\}$S2 \ {v} is a spanning set of VV .
- [ ] For any $v \in V \setminus S_1$ v ∈ V \ S1, $S_1 \cup \{v\}$S1 ∪ {v} is a linearly dependent.
- [ ] For any $v \in V$ v ∈ V , $S_2 \cup \{v\}$S2 ∪ {v} is a spanning set of VV .

No, the answer is incorrect.

Score: 0

Accepted Answers:

For any $v \in S_1$ v ∈ S1, $S_1 \setminus \{v\}$ S1 \ {v} is a linearly independent.

For any $v \in V \setminus S_1$ v ∈ V \ S1, $S_1 \cup \{v\}$ S1 ∪ {v} is a linearly dependent.

For any $v \in V$ v ∈ V , $S_2 \cup \{v\}$ S2 ∪ {v} is a spanning set of V V .

1 point

Consider the following subset of R^2 R2 with usual addition and scalar multiplication as in R^2 R2.

- $V_1 = \{(x, 0) \mid x \in R\}$ V1 = {(x, 0) | x ∈ R}
- $V_2 = \{(0, y) \mid y \in R\}$ V2 = {(0, y) | y ∈ R}

Which of the following options are correct?

- [ ] Both V_1 V1 and V_2 V2 are subspaces of R^2 R2.
- [ ] $V_1 \cap V_2$ V1 ∩ V2 is a subspace of R^2 R2.
- [ ] $V_1 \cup V_2$ V1 ∪ V2 is a subspace of R^2 R2.
- [ ] V_1 V1 is a subspace of R^2 R2, but V_2 V2 is not.
- [ ] V_2 V2 is a subspace of R^2 R2, but V_1 V1 is not.
- [ ] $\{(1, 0)\}$ {(1, 0)} is a basis of V_1 V1.
- [ ] $\{(0, 1)\}$ {(0, 1)} is a basis of V_2 V2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both V_1 V1 and V_2 V2 are subspaces of R^2 R2.

$V_1 \cap V_2$ V1 ∩ V2 is a subspace of R^2 R2.

$\{(1, 0)\}$ {(1, 0)} is a basis of V_1 V1.

$\{(0, 1)\}$ {(0, 1)} is a basis of V_2 V2.

1 point

Consider the following subset of R^2 R2 with usual addition and scalar multiplication as in R^2 R2.

- $V_1 = \{(x, 0) \mid x \in R\}$ V1 = {(x, 0) | x ∈ R}
- $V_2 = \{(2x, 0) \mid x \in R\}$ V2 = {(2x, 0) | x ∈ R}

Which of the following options are correct?

- [ ] Both V_1 V1 and V_2 V2 are subspaces of R^2 R2.
- [ ] $V_1 \cap V_2$ V1 ∩ V2 is a subspace of R^2 R2.
- [ ] $V_1 \cup V_2$ V1 ∪ V2 is a subspace of R^2 R2.
- [ ] V_1 V1 is a subspace of R^2 R2, but V_2 V2 is not.
- [ ] V_2 V2 is a subspace of R^2 R2, but V_1 V1 is not.
- [ ] $\{(1, 0)\}$ {(1, 0)} is a basis of V_1 V1.
- [ ] $\{(1, 0)\}$ {(1, 0)} is a basis of V_2 V2.

No, the answer is incorrect.

Score: 0

Accepted Answers:
Both V_1 V1 and V_2 V2 are subspaces of R^2 R2.
$V_1 \cap V_2$ V1 ∩ V2 is a subspace of R^2 R2.
$V_1 \cup V_2$ V1 ∪ V2 is a subspace of R^2 R2.
{(1, 0)}{(1, 0)} is a basis of V_1 V1.
{(1, 0)}{(1, 0)} is a basis of V_2 V2.

1 point

Let V V be a vector space which is defined as follows:

$$V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq R^4 \text{ V } = \{(x, y, z, w) \mid x + z = y + w\} \subseteq R4$$

with usual addition and scalar multiplication. Which of the following set forms a basis of V V ?

- [ ] {(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)}{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)}.
- [ ] {(1, 1, 0, 0), (0, 1, − 1, 0), (0, − 1, 0, 1)}{(1, 1, 0, 0), (0, 1, −1, 0), (0, −1, 0, 1)}.
- [ ] {(1, 0, − 1, 0), (1, 1, 0, 0), (1, 0, 0, 1)}{(1, 0, −1, 0), (1, 1, 0, 0), (1, 0, 0, 1)}.
- [ ] {(1, 1, 0, 0), (0, 1, 1, 0), (0, − 1, 0, 1)}{(1, 1, 0, 0), (0, 1, 1, 0), (0, −1, 0, 1)}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
{(1, 0, − 1, 0), (1, 1, 0, 0), (1, 0, 0, 1)}{(1, 0, −1, 0), (1, 1, 0, 0), (1, 0, 0, 1)}.
{(1, 1, 0, 0), (0, 1, 1, 0), (0, − 1, 0, 1)}{(1, 1, 0, 0), (0, 1, 1, 0), (0, −1, 0, 1)}.

Level 2

1 point

Which of the following sets form a basis of the vector space of $2 \times 22 \times 2$ lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct)

- [ ] $\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}\{[1000], [0001], [0100]\}$
- [ ] $\{\begin{bmatrix}1&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\1&1\end{bmatrix}, \begin{bmatrix}1&0\\0&1\end{bmatrix}\}\{[1100], [0101], [1001]\}$
- [ ] $\{\begin{bmatrix}0&0\\1&0\end{bmatrix}\}\{[0100]\}$
- [ ] $\{\begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}\{[-1000], [000-1], [0100]\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}\{[1000], [0001], [0100]\}$

$\{\begin{bmatrix}1&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\1&1\end{bmatrix}, \begin{bmatrix}1&0\\0&1\end{bmatrix}\}\{[1100], [0101], [1001]\}$

$\{\begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}\{[-1000], [000-1], [0100]\}$

If SS is a linearly independent set in vector space $V = \{(x, y) \mid y = 2x,$ where $x, y \in R\}$V = {(x, y) | y = 2x, where x, y ∈ R} with usual addition and scalar multiplication as in R^2 R2, then find the maximum possible cardinality of SS.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
If $\{v_1, v_2, v_3\}${v1, v2, v3} forms a basis of R^3 R3, then which of the following are true?

- [ ] $\{v_1, v_2, v_1 + v_3\}${v1, v2, v1 + v3} forms a basis of R^3 R3
- [ ] $\{v_1, v_1 + v_2, v_1 + v_3\}${v1, v1 + v2, v1 + v3} forms a basis of R^3 R3.
- [ ] $\{v_1, v_1 + v_2, v_1 - v_3\}${v1, v1 + v2, v1 − v3} forms a basis of R^3 R3.
- [ ] $\{v_1, v_1 - v_2, v_1 - v_3\}${v1, v1 − v2, v1 − v3} forms a basis of R^3 R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{v_1, v_2, v_1 + v_3\}${v1, v2, v1 + v3} forms a basis of R^3 R3
$\{v_1, v_1 + v_2, v_1 + v_3\}${v1, v1 + v2, v1 + v3} forms a basis of R^3 R3.
$\{v_1, v_1 + v_2, v_1 - v_3\}${v1, v1 + v2, v1 − v3} forms a basis of R^3 R3.
$\{v_1, v_1 - v_2, v_1 - v_3\}${v1, v1 − v2, v1 − v3} forms a basis of R^3 R3.
1 point
Which of the following options is(are) true?

- [ ] Any minimal spanning set of a vector space VV must be a basis of VV .
- [ ] Any maximal spanning set of a vector space VV must be a basis of VV .
- [ ] Any minimal linear independent set of vector space VV must be a basis of VV .
- [ ] Any maximal linear independent set of vector space VV must be a basis of VV .
- [ ] The basis of a vector space is unique.
- [ ] The number of elements in a basis of R^3 R3 is 33.
- [ ] The number of elements in a basis of $M_{3\times 3}(R)$M3×3(R) is 33.
- [ ] There are infinite number of bases of R^3 R3.
- [ ] Any subset of a minimal spanning set of VV cannot be a spanning set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Any minimal spanning set of a vector space VV must be a basis of VV .
Any maximal linear independent set of vector space VV must be a basis of VV .
The number of elements in a basis of R^3 R3 is 33.

There are infinite number of bases of R^3 R3.
Any subset of a minimal spanning set of V V cannot be a spanning set.

Check Answers
Your score is: 0/10